به نام خدا

استاد: خانم رضایی | آزمایشگاه سیستم دیجیتال ۲ | علی سپهرنیا (۹۶۱۲۱۰۳۸)

پاسخ تمرین های جلسه سوم

تمرین ۱:

برنامه ای بنویسید که در سطر اول نمایشگر نام شما از سمت چپ LCD به صورت حرف به حرف ظاهر شود و سپس از حرف آخر شروع به حذف شدن کند.

G:\Personal\Uni\Micro Lab\Micro\Az3_Tamrin\Az3_Tamrin.c

Notes | Az3_Tamrin.c

```
/*****************************************************
This program was created by the
CodeWizardAVR V3.12 Advanced
Automatic Program Generator
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Project :
Version :
Date    : 10/8/2020
Author  : Ali Sepehrnia (96121038)
Company :
Comments:


Chip type               : ATmega16
Program type            : Application
AVR Core Clock frequency: 1.000000 MHz
Memory model            : Small
External RAM size       : 0
Data Stack size         : 256
*****************************************************/

#include <mega16.h>
#include <delay.h>

// Alphanumeric LCD functions
#include <alcd.h>

// Declare your global variables here
int i,j;

void main(void)
{
// Declare your local variables here
```

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Notes | Az3_Tamrin.c

```
char name[] = "Ali Sepehrnia";

// Input/Output Ports initialization
// Port A initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRA=(0<<DDA7) | (0<<DDA6) | (0<<DDA5) | (0<<DDA4) | (0<<DDA3) | (0<<DDA2) | (0<<DDA1) | (0<<DDA0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTA=(0<<PORTA7) | (0<<PORTA6) | (0<<PORTA5) | (0<<PORTA4) | (0<<PORTA3) | (0<<PORTA2) | (0<<PORTA1) | (0<<PORTA0);

// Port B initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRB=(0<<DDB7) | (0<<DDB6) | (0<<DDB5) | (0<<DDB4) | (0<<DDB3) | (0<<DDB2) | (0<<DDB1) | (0<<DDB0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);

// Port C initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRC=(0<<DDC7) | (0<<DDC6) | (0<<DDC5) | (0<<DDC4) | (0<<DDC3) | (0<<DDC2) | (0<<DDC1) | (0<<DDC0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTC=(0<<PORTC7) | (0<<PORTC6) | (0<<PORTC5) | (0<<PORTC4) | (0<<PORTC3) | (0<<PORTC2) | (0<<PORTC1) | (0<<PORTC0);

// Port D initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRD=(0<<DDD7) | (0<<DDD6) | (0<<DDD5) | (0<<DDD4) | (0<<DDD3) | (0<<DDD2) | (0<<DDD1) | (0<<DDD0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) | (0<<PORTD1) | (0<<PORTD0);

// Alphanumeric LCD initialization
// Connections are specified in the
// Project|Configure|C Compiler|Libraries|Alphanumeric LCD menu:
// RS - PORTA Bit 0
// RD - PORTA Bit 1
// EN - PORTA Bit 2
// D4 - PORTA Bit 4
// D5 - PORTA Bit 5
```

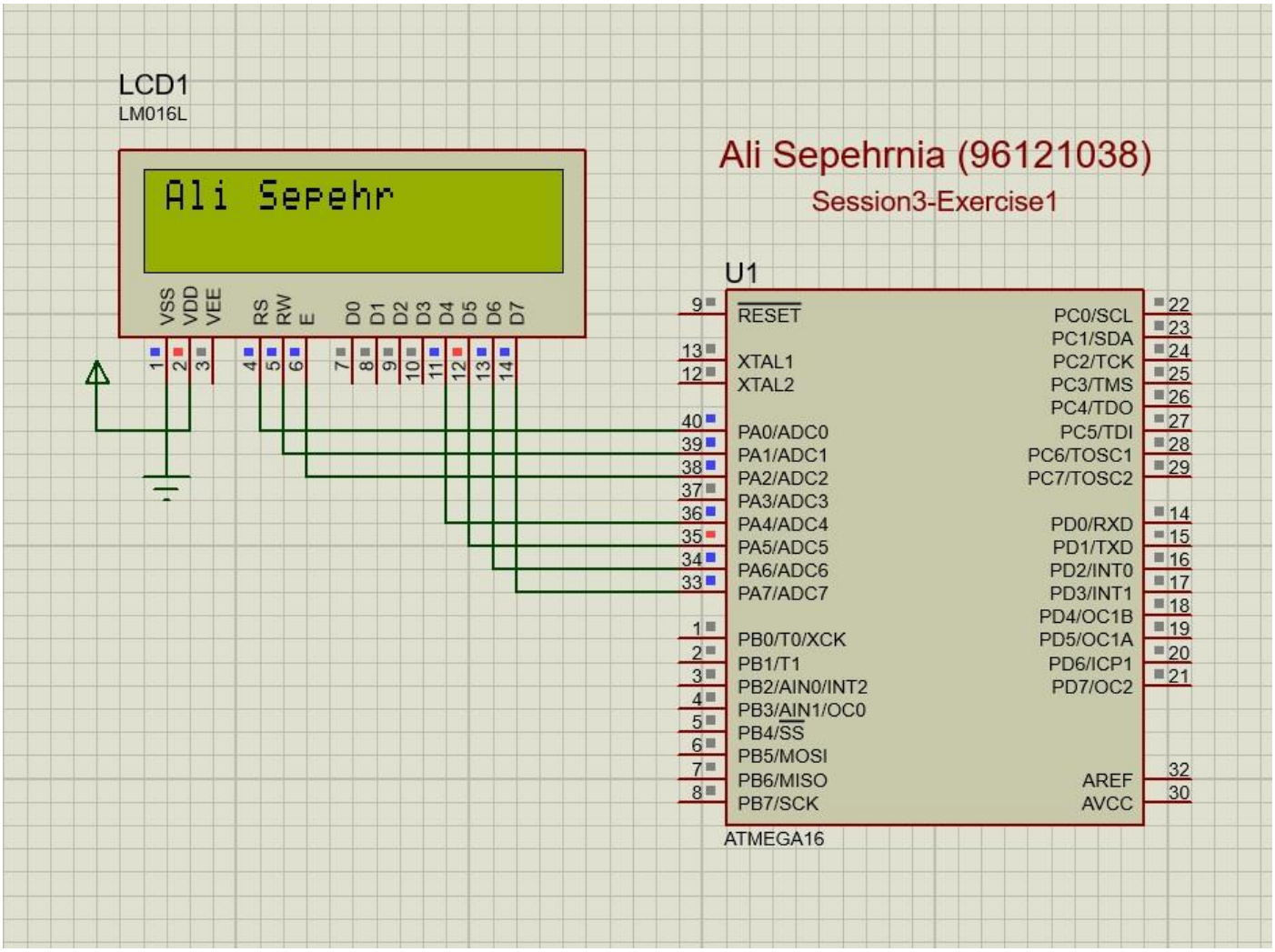
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Notes | Az3_Tamrin.c

```
// D5 - PORTA Bit 5
// D6 - PORTA Bit 6
// D7 - PORTA Bit 7
// Characters/line: 16
lcd_init(16);

while (1)
      {
      delay_ms(500); //Micro Restart Delay

      for(i=0;i<sizeof(name);i++) //Insert Letters By Letters
      {
          lcd_gotoxy(i,0);
          lcd_putchar(name[i]);
          delay_ms(250); //Animation Frame Speed
      }
      for(i=sizeof(name)-1;i>=0;i--) //Remove Letters By Letters
      {
          lcd_clear(); //Clear Screen From Last Remainded Characters
          for(j=0;j<i;j++) //Show Substring of name from name[0] to name[j]
          {
              lcd_gotoxy(j,0);
              lcd_putchar(name[j]);
          }
          delay_ms(250); //Animation Frame Speed
      }
      }
}
```



تمرین ۲:

برنامه ای بنویسید که در سطر دوم نمایشگر، نام شما به صورت تابلو روان از چپ به راست حرکت کند.

G:\Personal\Uni\Micro Lab\Micro\Az3_Tamrin2\Az3_Tamrin2.c

Notes | Az3_Tamrin2.c

```
1  /*****************************************************
2  This program was created by the
3  CodeWizardAVR V3.12 Advanced
4  Automatic Program Generator
5  © Copyright 1998-2014 Pavel Haiduc, HP InfoTech s.r.l.
6  http://www.hpinfotech.com
7
8  Project : Session3-Exercise2
9  Version :
10 Date    : 10/9/2020
11 Author  : Ali Sepehrnia (96121038)
12 Company :
13 Comments:
14
15
16 Chip type               : ATmega16
17 Program type            : Application
18 AVR Core Clock frequency: 1.000000 MHz
19 Memory model            : Small
20 External RAM size       : 0
21 Data Stack size         : 256
22 *****************************************************/
23
24 #include <mega16.h>
25 #include <delay.h>
26
27 // Alphanumeric LCD functions
28 #include <alcd.h>
29
30 // Declare your global variables here
31
32 void main(void)
33 {
34 // Declare your local variables here
35 char name[] = "Ali Sepehrnia";
```

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Notes | Az3_Tamrin2.c

```
35 char name[] = "Ali Sepehrnia";
36 int length = sizeof(name)-1; //-1 Is For That Strings End With An Empty Character
37 int i,j;
38
39 // Input/Output Ports initialization
40 // Port A initialization
41 // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
42 DDRA=(0<<DDA7) | (0<<DDA6) | (0<<DDA5) | (0<<DDA4) | (0<<DDA3) | (0<<DDA2) | (0<<DDA1) | (0<<DDA0);
43 // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
44 PORTA=(0<<PORTA7) | (0<<PORTA6) | (0<<PORTA5) | (0<<PORTA4) | (0<<PORTA3) | (0<<PORTA2) | (0<<PORTA1) | (0<<PORTA0);
45
46 // Port B initialization
47 // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
48 DDRB=(0<<DDB7) | (0<<DDB6) | (0<<DDB5) | (0<<DDB4) | (0<<DDB3) | (0<<DDB2) | (0<<DDB1) | (0<<DDB0);
49 // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
50 PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);
51
52 // Port C initialization
53 // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
54 DDRC=(0<<DDC7) | (0<<DDC6) | (0<<DDC5) | (0<<DDC4) | (0<<DDC3) | (0<<DDC2) | (0<<DDC1) | (0<<DDC0);
55 // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
56 PORTC=(0<<PORTC7) | (0<<PORTC6) | (0<<PORTC5) | (0<<PORTC4) | (0<<PORTC3) | (0<<PORTC2) | (0<<PORTC1) | (0<<PORTC0);
57
58 // Port D initialization
59 // Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
60 DDRD=(0<<DDD7) | (0<<DDD6) | (0<<DDD5) | (0<<DDD4) | (0<<DDD3) | (0<<DDD2) | (0<<DDD1) | (0<<DDD0);
61 // State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
62 PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) | (0<<PORTD1) | (0<<PORTD0);
63
64 // Alphanumeric LCD initialization
65 // Connections are specified in the
66 // Project|Configure|C Compiler|Libraries|Alphanumeric LCD menu:
67 // RS - PORTA Bit 0
68 // RD - PORTA Bit 1
69 // EN - PORTA Bit 2
```

Line: 35

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Notes | Az3_Tamrin2.c

```
lcd_init(16);


while (1)
      {
      delay_ms(150); //Micro Restart Delay

      for(i=-length+1;i<16;i++) //Animation Loop
      {
            lcd_clear();
            lcd_gotoxy(i,1);

            if(i<0) //For Coming From The Left Side
            {
                lcd_gotoxy(0,1);
                for(j=-i;j<length;j++)
                {
                    lcd_putchar(name[j]);
                }
            }
            else if(i>16-length) //For Exiting From The Right Side
            {
                for(j=0;j<=16-i;j++)
                {
                    lcd_putchar(name[j]);
                }
            }
            else
            {
                lcd_puts(name);
            }

            delay_ms(150);
      }
      lcd_clear(); //Clear Screen After Animation
      }
```

